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Cold atmospheric plasma as a testable redox input in plants: Chemistry, biological dose and evidence requirements

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ABSTRACT

Cold atmospheric plasma (CAP) and plasma-activated water (PAW) provide chemically diverse inputs that can modify plant establishment, physiology, and defense. Their effects remain difficult to transfer because exposure duration does not specify the chemistry reaching responsive tissues. This critical narrative review develops redox programming as a testable input–response framework, rather than an established precision technology. We connect plasma generation and post-discharge chemistry with tissue accessibility, endogenous buffering, calcium dynamics, protein redox regulation, and agricultural outcomes. A biological redox dose is operationalized as a multidimensional exposure-and-context descriptor with separately reported chemical amounts, concentration–time integrals, delivery geometry, and plant-state measurements. A published wheat dataset illustrates both its measurable components and the information that cannot be reconstructed retrospectively. The signaling window is explicitly related to established oxidative-window and hormesis concepts; the added contribution is a measurement and causal-validation workflow, not a new universal dose–response law. Selected primary studies are compared using mechanistic evidence grades and separate application-readiness categories. Chemistry-matched comparisons show why nitrate nutrition, acidity and physical surface modification must be distinguished from redox signaling. We specify analytical controls, plant-compatible biosensors, and a crop-specific optimization example. CAP is therefore a promising experimental redox input, but its programmability requires prospective chemical, mechanistic, agronomic, and economic validation.

1. Introduction: from oxidative stress to engineered redox signaling

Reactive oxygen and nitrogen species in plants are more than toxic by-products; their identity, location, concentration, and lifespan let them convey information about development and environmental change [1]. Substances such as nitrite, nitrate, hydrogen peroxide, and other stable compounds in plasma-activated water extend this chemistry beyond mere discharges, providing a mobile input for agricultural practices [2]. The formation pathways and physicochemical properties of plasma-activated water depend on factors such as reactor configuration, feed gas, liquid composition, and post-treatment storage conditions. Consequently, similar liquids may not produce identical biological effects [3]. Reactive species can target biomolecules selectively or broadly, depending on their lifetimes and reaction kinetics, so their composition and flux are more biologically meaningful than the general term "oxidative stress" [4]. Conventional crop-stress interventions typically influence water, mineral, hormone, or microbial conditions and only indirectly modify the intracellular redox state. CAP is distinctive because it can rapidly induce a chemically diverse redox disturbance without requiring a stable transgene or an ongoing synthetic regulator [5]. This targeted use of redox disturbance is termed redox programming: deliberately controlling redox conditions and redox-sensitive signaling to steer an inherent response toward a specific phenotype. This term does not imply deterministic control, heritable genome editing, or permanent changes; instead, it refers to an input–response approach whose effectiveness must be verified through chemical, mechanistic, and phenotypic evaluation. Outcomes vary with source, treatment regime, and seed state rather than following a uniformly beneficial pattern [6,7]. The decisive scientific problem is therefore not whether CAP "produces ROS," but whether a defined external flux becomes an organized signal before antioxidant and repair systems are overwhelmed. A useful review must connect four levels that are too often treated separately: plasma generation, interfacial delivery,

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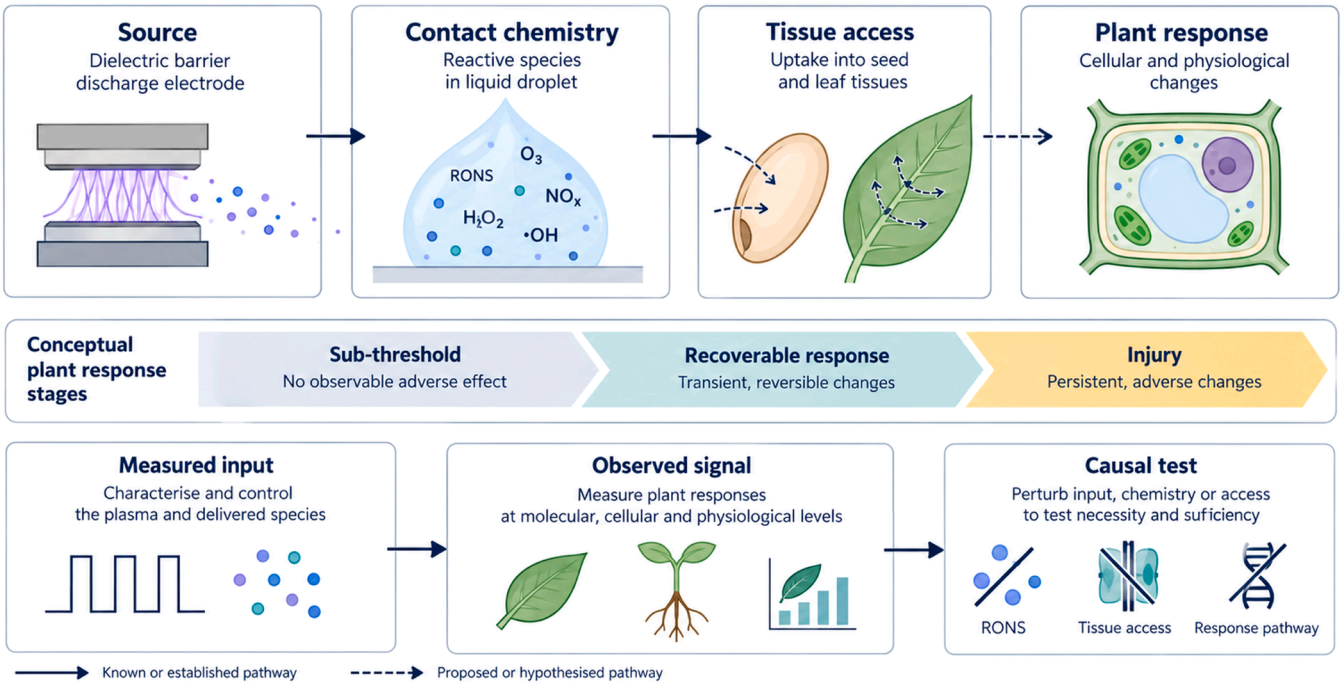


Fig. 1. Measured input and testable plant response. An original conceptual synthesis linking source settings, measured contact chemistry, tissue access, and plant-state-dependent decoding to a prospective phenotype. The separate evidence strip distinguishes measured exposure, observed response, and causal validation.

endogenous signal processing, and agricultural performance. We consequently ask three questions: what is delivered to the plant, how is that input decoded, and what evidence is required before a treatment can be called programmable?

The review follows an explicit source-to-response structure. Section 2 introduces CAP, device-dependent chemistry, and the operational dose descriptor. Section 3 follows delivery through the plant interface and defines the response window. Sections 4–6 examine molecular decoding, hormonal interactions and metabolic consequences. Section 7 distinguishes stimulation, priming, and memory. Section 8 compares agricultural evidence and chemistry-matched controls; Section 9 specifies plasma-focused omics designs. Section 10 presents local optimization, Section 11 defines reporting and evidence requirements, and Section 12 states the resulting conclusions. Mechanisms established in general plant biology are distinguished throughout from mechanisms demonstrated under CAP or PAW.

This is a critical narrative synthesis, not a systematic review or a new meta-analysis. The selected published literature provides the starting corpus, with targeted verification of chemical mechanisms, pivotal control experiments, and reference inconsistencies. Table 5 presents a deliberately selected evidence matrix; its denominator should not be interpreted as all published plasma–plant papers. Mechanistic grades express the strength of the redox claim, whereas application readiness expresses the extent of production validation. This distinction allows a crop-cycle phenotype to have relatively advanced agronomic follow-up while remaining mechanistically unresolved. (Fig. 1)

Table 1
Plant-relevant plasma modalities and delivery conditions.

Modality	Controls and chemistry	Plant relevance and main caveat	Reference
Direct DBD	Waveform, power density, gap, humidity; surface RONS and ozone/NOx	Seed treatment; packing and heat/field/UV effects require controls	[6,7]
Atmospheric jet	Gas flow, entrainment, nozzle distance; spatially varying effluent	Local seed or leaf exposure; footprint and gradients limit transfer	[8,10]
Corona or spark	Polarity, current, gap and residence time; source-specific aqueous products	Water activation or surface contact; discharge regime must be identified	[11,12]
Batch PAW	Power, exposed volume, depth, mixing; H_2O_2, nitrite, nitrate and pH at use	Soaking, irrigation or foliar delivery; storage changes chemistry	[13,14]
Flow-through PAW	Flow, residence-time distribution, gas–liquid transfer and energy per liter	Repeated or larger-volume water treatment; quality needs inline verification	[15,16]
Activated nutrient medium	Initial salts, buffers, organics and post-treatment concentrations	Hydroponics/aeroponics; nutrient and redox effects must be separated	[17,18]

2. Cold atmospheric plasma as a biological redox generator

2.1. Plasma systems relevant to plants

Plant experiments typically use direct irradiation, plasma-activated liquids, and plasma-treated soil as delivery methods [8]. These categories describe broad approaches rather than specific biological exposures. Studies comparing soybean treatments with three atmospheric-pressure sources showed that researchers must assess both source traits and seed responses together [6]. Plasma systems that activate liquids can be used for irrigation, hydroponics, and foliar applications after activation, although the stored liquids lose most of the short-lived reactive species present during discharge [9]. The key unit of comparison is an experimentally characterized treatment chain, including electrical input, gas phase, interface, target, and contact time before biological interaction. Without details such as voltage waveform, absorbed power, gas flow, humidity, nozzle distance, treated area or volume, and temperature, the device's reproducibility is limited. Conversely, detailed plasma physics offers minimal value if the parameters reaching the biological target are unmeasured. Therefore, Table 1

categorizes systems based on exposure-relevant properties that should accompany biological data.

CAP is a partially ionized gas generated at atmospheric pressure, where energetic electrons drive excitation, dissociation, and ionization while the bulk gas remains comparatively cool. "Cold" describes nonequilibrium operation; it does not establish the temperature or safety of a particular seed or leaf. Dielectric-barrier discharges limit current through an insulating barrier, jets transport plasma effluent toward a local target, and corona or spark configurations concentrate discharge activity near electrodes. Electrons initiate chemistry, but the target can also receive photons, fields, charged particles, gas products, heat, and RONS [8,6,19].

Feed gas and ambient entrainment determine available oxygen and nitrogen chemistry. Humidity changes water-derived radical formation; absorbed power and waveform change electron-driven reactions; distance and gas residence time change transport and loss. Liquid depth, mixing, and exposed area set the gas–liquid transfer conditions. Bicarbonate, chloride, transition metals and organic carbon redirect subsequent aqueous reactions. Consequently, changing voltage or activation time can change both concentration and composition. A device comparison is meaningful only when the measured chemistry, delivery conditions and target properties are compared together (Table 1) [13,20].

2.2. Reactive-species generation, transport, and persistence

Gas-phase reactions and subsequent water chemistry can produce measurable oxygen- and nitrogen-based species, with yields influenced by the plasma operating conditions [21]. Microbubble-assisted plasma water production demonstrates how gas–liquid mass transfer can modify nitrogen fixation and affect plant growth [15]. The target also actively participates in the chemistry: salts, organic matter, seed-coat compounds, buffers, and exudates either consume or transform the incoming species. Thus, 'plasma composition' is not a fixed source property but rather results from source chemistry, transport processes, interfacial reactions, and target demand. Short-lived radicals react quickly locally but penetrate only briefly, while longer-lived species can diffuse, persist, and enable delayed signaling. Although useful, this division is incomplete because secondary products can be regenerated near membranes, and co-occurring species may react synergistically. Reactive nitrogen chemistry is particularly sensitive to pH and oxygen levels, whereas hydrogen peroxide stability depends on catalysts, light, temperature, and organic load. Therefore, a biologically relevant exposure description should include both immediate and time-resolved chemistry, rather than relying on a single end-point concentration.

Post-discharge PAW remains a reacting solution. An acid-dependent overall pathway is $H_2O_2 + NO_2^- + H^+ \rightarrow ONOOH + H_2O$. This is a net reaction description, not a claim that the written equation is an elementary step. Nitrous acid dissociates into H^+ and NO_2^- with an approximate pK_a near 3.3; thus, acidification substantially increases the protonated nitrite fraction. The $ONOOH/ONOO^-$ pair has a pK_a near 6.8. Under acidic aqueous conditions, ONOOH is transient, with decay on a subsecond-to-second timescale depending on temperature and matrix. Its short lifetime does not exclude continuing formation while precursor peroxide and nitrite coexist [12].

A storage-time measurement must therefore distinguish the intermediate's lifetime from the lifetime of the mixture that generates it. Peroxide and nitrite can be consumed after discharge stops, while nitrate can accumulate as a relatively persistent oxidized-nitrogen product. Ozone loss, gas exchange, light exposure, vessel surfaces, and trace metals can change the trajectory. Cooling may slow some reactions but cannot replace measurements at the time of plant contact. Record concentrations immediately after generation, before application, and during representative target contact, with a target-free storage control. Neutralization changes speciation and reaction rates and should be followed by renewed chemical characterization.

Table 2
Illustrative PAW compositions reported in plant studies.

Plant formulation	H_2O_2	Nitrite and nitrate	Reference
Mung bean air/N_2 PAW	52.68–63.79 μM	NO_2^-: 2.12–2.64 mg L^{-1} (46.1–57.4 μM); NO_3^-: 5.55–22.16 mg L^{-1} (89.5–357.4 μM)	[14]
Wheat PAW	97.4 mg L^{-1} (2.86 mM)	NO_2^- <0.001 mg L^{-1}; NO_3^- 391.1 mg L^{-1} (6.31 mM)	[11]
Lettuce diluted PAW	Not used here for a cross-study peroxide estimate	NO_3^-: 1.5–3.0 mg L^{-1} (24.2–48.4 μM)	[22]
Tomato nutrient-replacement PAW	Not used here for a cross-study peroxide estimate	NO_3^-: 400 and 1000 mg L^{-1} (6.45 and 16.13 mM); pH 5.6–5.8	[18]

Values describe the cited formulations at their reported sampling conditions. They are not universal treatment thresholds, and unreported values are not zero. Mass-to-molar conversions use molecular masses of 34.01, 46.01 and 62.00 g mol^{-1} for peroxide, nitrite and nitrate, respectively.

Table 3
Operational quantities for biological redox dose.

BRD component	Measurable quantity and unit	Routine minimum	Specialist extension
C Chemical input	H_2O_2, NO_2^-, NO_3^- in μM or mM; n_i in μmol batch^{-1}, seed^{-1} or g^{-1}	Calibrated quantitative assays; applied volume and target mass	Speciation, isotope balance and interfacial products
C Acidity and matrix	pH; conductivity in μS cm^{-1}; temperature in °C; ORP in mV with reference	At generation and at use; matrix blanks	Buffer capacity titration; dissolved gas and metals
K Contact kinetics	Contact duration and storage delay in min; E_i in μM min	Clocked protocol; before/after contact chemistry	Repeated sampling to estimate $\int C_i$, target(t)dt
K Direct flux	J_i in nmol cm^{-2} s^{-1}; F_i in nmol cm^{-2}	Do not infer species flux from treatment time	Species-specific diagnostics with spatial calibration
A Geometry and access	Treated area in cm^2; liquid/seed ratio in mL g^{-1}; imbibition in g water g^{-1}	Area, volume, seed packing, moisture and surface wetting	Contact angle, microscopy, transport and tissue mapping
B Baseline pools	GSH/GSSG and ascorbate/dehydroascorbate in nmol g^{-1} fresh mass; ratios	Pretreatment matched biological samples	Compartment-specific redox reporters; E_GSH in mV
B Plant context	Genotype, lot, stage, nutrient input, prior stress and time of day	Metadata with independent biological units	Validated crop-specific buffer-state model
Separate response validation	Reporter peak ratio, integrated response in ratio min, recovery in min	Injury and phenotype with recovery time course	Live-cell kinetics and site-resolved protein redox state

Reported agricultural formulations span substantially different chemical scales. Lettuce received 1.5–3.0 mg L^{-1} nitrate [22], whereas tomato nutrient-replacement treatments used 400 and 1000 mg L^{-1} nitrate after pH adjustment to 5.6–5.8 [18]. Table 2 includes a peroxide-rich wheat formulation and a nitrite reporting limit. These are study-specific examples, not a pooled "typical PAW" distribution. A universal concentration range would obscure differences in reactor, dilution, storage, and nutrition; the available examples span low-micromolar to millimolar nitrate. Nitrite below a method's detection limit is a censored measurement, not zero.

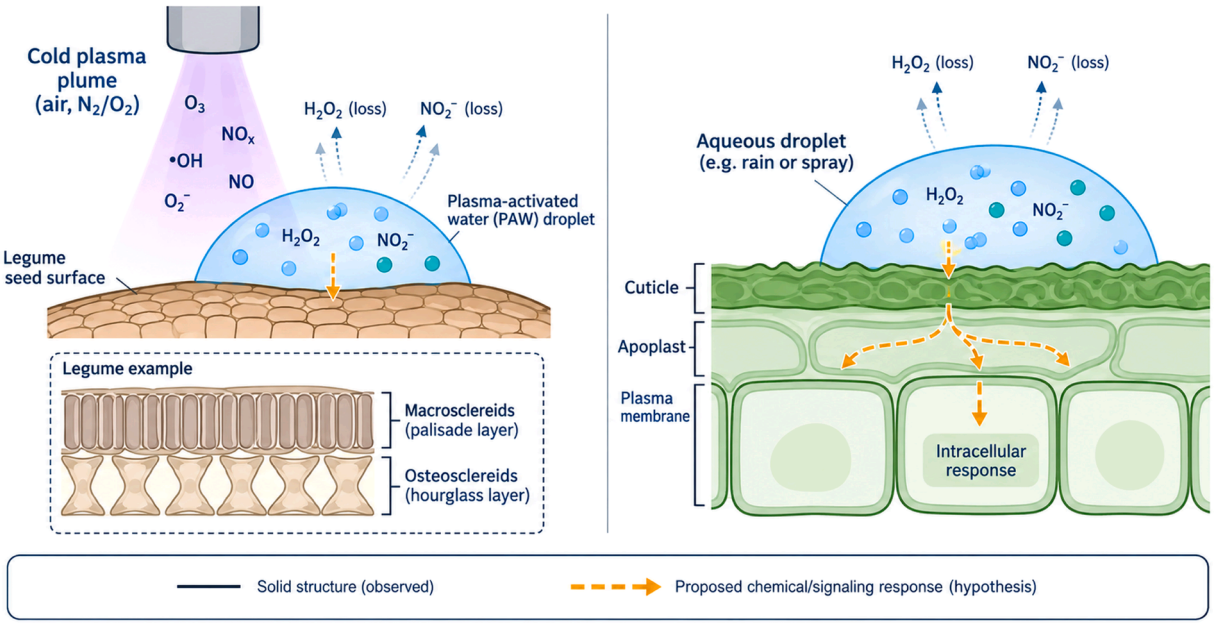


Fig. 2. Chemistry at the plasma–plant interface. An original schematic of PAW or direct-plasma contact with the seed or leaf surface, followed by apoplastic conversion and membrane-associated signaling. The legume seed-coat inset is illustrative: macrosclereids form the palisade layer and osteosclereids the hourglass layer. Seed-coat anatomy varies among taxa. Transient radicals are shown at the interface; long-lived precursor species are shown in water.

2.3. The plasma dose problem and a biological redox dose

We define BRD = (C, K, A, B), where C is species-resolved external chemical exposure, K is its time course, A is target accessibility and delivery geometry, and B is the pretreatment biological buffering state. BRD is an exposure-and-context descriptor, not a directly measured intracellular dose or a universal scalar. The adjective biological refers to the required target context, not to an assumed conversion of applied molecules into absorbed redox equivalents. Nitrate, acidity, and energy are covariates with different meanings and units; they must not be added to peroxide concentration.

For each stable species i, report the applied amount $n_i = \Sigma_j C_{i,j}V_j$ in μmol per batch, seed, or gram of dry seed, retaining the measured concentration $C_{i,j}$ in μmol L^{-1} and application volume V_j in liters. Where contact kinetics are sampled, report $E_i = \int_0^T C_i$, target(t)dt in μmol L^{-1} min. For direct exposure, a measured species flux J_i can be integrated as $F_i = \int J_i dt$ in nmol cm^{-2}. These quantities describe different experiments and are not interchangeable. Loss from the bathing solution includes abiotic decay, surface adsorption, microbial demand, and plant uptake; without a mass balance and appropriate controls, it cannot be called absorption (Table 3).

Routine measurements include calibrated bulk peroxide, nitrite, and nitrate assays; pH; conductivity; liquid volume; contact duration; temperature; seed mass; and moisture. Time-resolved surface flux, apoplastic chemistry and compartmental reporter trajectories require specialized methods. A minimal BRD record is still useful when advanced components are unavailable, provided missing fields are explicitly reported as unmeasured. Baseline glutathione and ascorbate pools, their oxidized fractions and antioxidant activities characterize B; they do not justify a single "buffer capacity" coefficient unless the coefficient is validated for a specific assay and tissue.

Worked example from Jirešová et al. [11]: H_2O_2 97.4 ± 3.2 mg L^{-1} = 2.86 ± 0.09 mM; nitrate 391.1 ± 9.3 mg L^{-1} = 6.31 ± 0.15 mM; nitrite <0.001 mg L^{-1} = <0.022 μM. The initial 3 mL for 30 wheat grains supplied 8.59 μmol peroxide and 18.92 μmol nitrate per dish, or 0.286 and 0.631 μmol per grain. These are nominal applied amounts. Daily replenishment was 1 mL. Contact concentration–time integrals, tissue uptake and baseline buffers were unmeasured; they cannot be reconstructed from device time. The nitrate-derived nitrogen content was 88.4 mg N L^{-1}. This example exposes the measurable and missing BRD components rather than retrospectively proving a biological dose. (Fig. 2)

3. The plasma–plant interface: translating external chemistry into a signal

3.1. Surface conversion rather than passive entry

The first biological target is usually a composite interface comprising wax, cuticle, seed-coat polymers, adherent microorganisms, water films, and dissolved solutes. Cold radiofrequency plasma has been shown to modify seed wettability and germination speed, supporting a surface-mediated route to altered imbibition [19]. Soybean studies combining seed-surface characterization, DNA analysis, germination, and plasma diagnostics show why researchers should evaluate surface modification and biological safety together [7]. Therefore, CAP edits the boundary conditions under which endogenous metabolism restarts. Cold-plasma modification of Lepidium perfoliatum seed gum demonstrates that plasma can alter both seed-derived surface polymers and intact seed coats [23]. Leaf exposure creates a different interface. Stomata, cuticle, trichomes, boundary-layer humidity, and leaf age determine where liquid or gas products accumulate. Long-lived species can enter the apoplast via aqueous pathways, whereas short-lived species are more likely to modify extracellular proteins, lipids, and cell-wall components at the point of contact. Because apoplastic peroxidases, transition metals, and pH transform the applied mixture, the receptor-facing signal may differ substantially from the species measured in the source.

For direct CAP, a factorial separation should include untreated handling, gas-flow sham, matched thermal exposure, and remote afterglow or PAW-only treatments. UV-transparent and UV-blocking barriers can probe photon contributions only after downstream gas chemistry and temperature are remeasured. A grounded mesh can change charged-particle and field exposure but also changes transport and reactive-species losses. Seed-coat etching should be quantified by microscopy, surface spectroscopy, contact angle, and imbibition kinetics. Mechanical scarification and matched oxidant treatments provide complementary

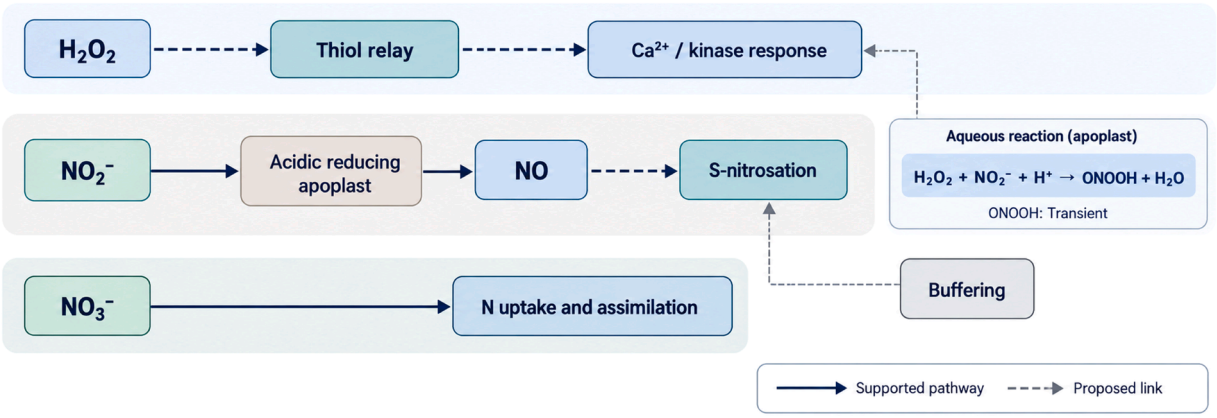


Fig. 3. Distinct chemical routes and candidate signaling. An original synthesis separates peroxide-dependent signaling, nitrite-derived NO and transient ONOOH chemistry from nitrate nutrition. Dashed arrows identify CAP-specific signaling hypotheses. No hydroxyl/superoxide interconversion or nitrate-to-excess-RONS arrow is implied.

comparators. No single barrier leaves every other plasma component unchanged; several orthogonal controls are required before assigning the phenotype solely to redox signaling [6,7,19].

3.2. From apoplast to cytosol

Plants already have machinery to convert extracellular redox changes into cellular information. Hydrogen peroxide-sensitive thiol systems can relay oxidation to target proteins, allowing a chemically simple molecule to generate pathway-selective consequences [24]. Thiol peroxidases combine peroxide removal with redox relay and metabolic tuning, making them plausible gatekeepers between CAP-derived peroxide and endogenous regulation [25]. Dynamic thiol proteomics shows that redox control is distributed across networks rather than confined to a small set of antioxidant enzymes [26]. Interactions among ROS, RNS, and phytohormones are strongly context-dependent during drought, which cautions against assigning a single meaning to an externally supplied species [27]. Spatial organization is equally important because redox signaling depends on production sites, scavenging domains, and intercellular propagation [28]. Calcium-regulated control of ROS influx provides a mechanism by which the plasma membrane can amplify or constrain a local oxidative perturbation [29]. In immune signaling, ROS production and mitogen-activated protein kinase activity show distinct dynamics, illustrating how timing, not mere presence, shapes downstream interpretation [30]. Jasmonate and salicylate can modulate systemic ROS-wave production in opposite directions, showing that hormonal state influences how a local redox event propagates [31].

3.3. A signaling-window model

We propose three operational regimes. In the sub-threshold regime, the external perturbation is absorbed by the surface and antioxidant network without a detectable regulatory response. In the programming window, the perturbation is large and well timed enough to alter calcium, protein redox state, kinase activity, transcription, or metabolism while preserving recovery. In the injury regime, membrane integrity, photosynthetic function, macromolecular quality, or growth fails to recover. These regimes should be diagnosed by trajectories, not by a single measurement immediately after treatment. A practical experiment should therefore pair early markers—surface chemistry, cytosolic calcium, apoplastic or organellar peroxide, protein oxidation, and kinase activation—with later markers of repair, growth, and performance. The central causal test is not whether ROS rose, but whether blocking or genetically altering a candidate sensor removes the downstream benefit without simply changing general stress tolerance. This approach turns a correlation between CAP and antioxidant enzymes into a falsifiable signaling mechanism.

The three regimes are consistent with the established oxidative window for seed germination and with hormetic stimulation at lower exposure and inhibition at higher exposure [32]. We do not claim novelty for a biphasic curve or for the signaling–damage distinction. Our contribution is the joint specification of species-resolved delivery, accessibility, pretreatment buffering, recovery trajectories and causal controls. A response window becomes transferable only if these measurements predict its location in a new batch; a device-time optimum alone remains a local empirical optimum.

External PAW concentrations cannot be equated with concentrations at a cytosolic sensor. Published plant reporters can resolve small peroxide-dependent changes, but their fluorescence reflects probe oxidation and reduction, compartment and imaging conditions, rather than a universal tissue concentration threshold [33]. Applied millimolar peroxide can therefore coexist with a much smaller, spatially restricted intracellular perturbation because of surface consumption, membrane transport, and scavenging. Nitrite must first be interpreted through pH-dependent speciation and local conversion, and nitrate requires a nutritional and signaling interpretation. There is no justified numerical "safe signaling concentration" common to seed coats, roots and leaves.

For scale, the HyPer7 study discusses approximate detection sensitivities of 30 nM for HyPer7 and 100 nM for roGFP2-Orp1 [33]. These are reporter-performance estimates, not physiological signaling thresholds or absolute intracellular concentrations after PAW. The contrast with the micromolar-to-millimolar applied formulations in Table 2 shows why we cannot assign a bath concentration directly to a cellular response window.

Operationally, the programming window requires a reproducible early reporter response above sham variation, subsequent return toward the pretreatment state, and an independently confirmed benefit without persistent injury. Predefine observation times and crop-specific acceptance criteria. Record peak reporter ratio, area under the baseline-subtracted response curve, and recovery half-time, together with electrolyte leakage, photosynthetic recovery, and abnormal-seedling frequency. These response variables validate the exposure descriptor; they should not be included in pretreatment BRD in a way that makes prediction circular. Statistical comparisons should use independent plant or batch units and report effect estimates with uncertainty.

4. Molecular mechanisms of CAP-induced redox signaling

4.1. Peroxide sensing, thiol switches, and redox relays

Quantitative thiol-redox proteomics has mapped hydrogen peroxide

treatment in Arabidopsis, providing a methodological basis for identifying candidate CAP-responsive cysteines rather than inferring them from total antioxidant activity [34]. Field studies of water stress show that hydrogen peroxide and ascorbate operate as a coupled regulatory system, so the meaning of a peroxide pulse depends on the pre-existing reductive buffer [35]. Thiol-redox proteomics of chloroplast stress responses shows that peroxide sensitivity is embedded in organelle-specific protein networks, arguing against a universal whole-plant dose [36]. The most plausible early decoding mechanism is a relay rather than indiscriminate oxidation (Fig. 3). A peroxidase or thiol peroxidase reacts quickly with peroxide, undergoes a defined cysteine modification, and transfers that redox equivalent to a partner protein. Direct oxidation of lower-reactivity targets may still occur when local peroxide increases, but relay architecture offers speed and selectivity within the programming window. CAP experiments should combine targeted redox proteomics, genetic perturbation of candidate relay proteins, and time-resolved imaging; enzyme-activity assays alone cannot identify the molecular switch.

Nitric oxide and related nitrogen species add another layer of control through S-nitrosation, nitration, and interactions with ROS-derived chemistry. Potassium responses integrate calcium and ROS/RNS signals during abiotic stress, showing that ion status can alter how an imposed redox perturbation propagates [37]. The green alga Chlamydomonas provides a tractable comparative system for studying ROS signaling and thiol regulation across the green lineage [38]. Low-oxygen signaling reveals extensive coordination between ROS and nitric oxide, making tissue oxygenation a likely modifier of CAP-derived nitrogen chemistry [39]. In maize, ROS-scavenging capacity participates in nitric oxide and hydrogen sulfide crosstalk during thermotolerance, indicating that reactive-species networks cannot be interpreted on a species-by-species basis [40].

Nitrite can generate NO nonenzymatically in an acidic apoplast when reducing substrates are available; plant tissues have been shown to support this route [41]. PAW-derived nitrite is therefore a potential precursor, not a direct quantitative measure of NO reaching a protein. Neutralization, apoplastic buffering, oxygen availability, and reductant supply can change its fate. Plant NO measurements should use orthogonal approaches and assay-specific controls; fluorescent dye accumulation alone is insufficient to assign the source or absolute amount of NO [42].

S-nitrosation modifies cysteine thiols and is governed by local nitrosating chemistry, transnitrosation, and denitrosation; it is not accurately represented as unrestricted direct addition of NO to every thiol. Tyrosine nitration is chemically and functionally distinct and can arise through peroxynitrite-derived or peroxidase/nitrite-dependent chemistry. A rise in nitrotyrosine is neither proof of a unique ONOOH source nor sufficient evidence of injury. CAP-specific hypotheses require site-resolved protein modification, abundance normalization, reaction controls and a functional perturbation connecting the modification to the phenotype. These are mechanistic predictions of the framework, not outcomes established for all CAP treatments.

4.2. Calcium and kinase dynamics

Plasma-derived species can change channel activity, membrane properties, extracellular redox state, and local pH, each of which could initiate calcium entry. Calcium-dependent protein kinases connect calcium signatures to immune outputs and offer testable nodes for CAP-induced signal routing [43]. Contemporary models of plant defense treat calcium signaling as a distributed network with channel, sensor, kinase, phosphatase, and organellar components [44]. MAPK, SUMO, and resistance-protein pathways can converge at transcriptional complexes, showing how a transient upstream event may be integrated into a coherent gene-regulatory response [45]. A treatment that raises bulk calcium without resolving amplitude, frequency, subcellular origin, or recovery therefore provides limited mechanistic information. Cold-plasma treatment has altered the activities of enzymes required for pea germination, illustrating a biochemical response that should be positioned downstream of early signaling rather than treated as the initiating event [46]. The critical experiment is temporal ordering. If CAP-derived chemistry is upstream, a measurable extracellular or apoplastic perturbation should precede cytosolic calcium, which should precede kinase activation and transcriptional change. Pharmacological inhibition can nominate steps, but channel mutants, kinase mutants, complementation, and orthogonal biosensors are required to separate pathway causality from generalized loss of vigor. A successful mechanistic chain must also predict the boundary at which stronger treatment ceases to increase signaling and begins to suppress recovery.

Suitable plant reporters include roGFP2-Orp1 for peroxide-linked oxidation, Grx1-roGFP2 for glutathione redox potential, and compartment-targeted HyPer7 for peroxide dynamics [33,47]. Calcium candidates include cytosolic GCaMP variants, the ratiometric Cameleon YC3.6 and aequorin, and a green-light-excitable FRET design that has been tested in plants [48]. Sensor choice must follow the compartment, time resolution, and available optics. roGFP2-Orp1 reports the balance of oxidation and endogenous reduction; Grx1-roGFP2 is not a peroxide concentration sensor. HyPer7 improves pH robustness, but imaging itself can produce oxidative responses, and full in situ reduction can be difficult [33].

For CAP experiments, acquire a pretreatment baseline, synchronize treatment and imaging, and use plasma-off reporter controls, non-reporter autofluorescence controls, and appropriate oxidized/reduced calibration. Plasma light can saturate detectors; image between pulses or after a characterized transfer delay. Chlorophyll fluorescence, pH shifts, movement, and photobleaching can alter ratios. Simultaneous green peroxide and calcium reporters may overlap spectrally and require separate lines or validated spectral separation. Non-model crops face limitations in transformation time, silencing, targeting, and tissue access. Transient expression or reporter model plants can test mechanisms, but each crop still needs independent chemical and phenotypic calibration.

4.3. Transcription-factor networks

Redox and kinase signals are transduced into gene expression through transcription factor families including WRKY, NAC, AP2/ERF, bZIP, and MYB. AP2/ERF factors participate broadly in abiotic stress responses and tolerance, yet family-wide involvement does not establish that a particular factor directly senses a CAP-induced redox event [49]. Arabidopsis ERF56 links nitrate availability to growth control, which is relevant when plasma-activated water simultaneously changes redox chemistry and nitrogen supply [50]. Eggplant SmERF1 positively regulates salt-stress performance, providing a tractable example of transcriptomic nomination followed by functional validation [51]. These examples define an evidence ladder for CAP studies. Differential expression is hypothesis-generating; promoter occupancy and reporter responses establish regulatory associations; loss- and gain-of-function establish necessity or sufficiency; and redox state or kinase dependence links the factor to the imposed stimulus. Most CAP datasets currently occupy the first rung. The next advance will come from testing compact causal modules, such as peroxide relay–calcium signature–MAPK–WRKY, rather than publishing ever-longer lists of differentially expressed genes.

4.4. Antioxidant buffering as adaptive control

Antioxidant induction should not automatically be interpreted as detoxification. Glutathione–peroxisome interactions coordinate organellar and cellular redox homeostasis, so a change in one compartment can redistribute rather than simply lower oxidative load [52]. Chloroplast glutathione peroxidases contribute to redox homeostasis through functions that extend beyond bulk peroxide removal [53]. Glutathione

Table 4
Plasma-specific molecular and physiological responses.

Plant and treatment	Measured response	Interpretation and next causal test	Reference
Arabidopsis leaf CAP	Rapid stress-related transcripts	Temporal association; test early sensors and pathway perturbation	[58]
Arabidopsis seed CAP	Early seedling transcriptome	Molecular association; match developmental stage and chemistry	[59]
Arabidopsis CAP-derived RONS	Redox and seedling growth endpoints	Connect external mixture to compartmental signals and recovery	[60]
Plant cells exposed to CAP species	Transient intracellular Ca^{2+} and H_2O_2	Early live-cell response; validate crop phenotype separately	[10]
Mung bean PAW	RNA sequencing and selected qPCR	Gas-dependent chemistry and molecular responses; perturb candidates	[14]
Sorghum PAW	Transcriptome and seedling biomass	Nutrition and redox remain coupled without nitrate-matched control	[61]
Barley PAW plus waterlogging	Leaf ascorbate	Defined buffer pool; use factorial recovery and rechallenge design	[62]
Basil direct CAP	Pathway transcripts and essential oils	Candidate metabolic response; quantify flux and dose-dependent injury	[63]

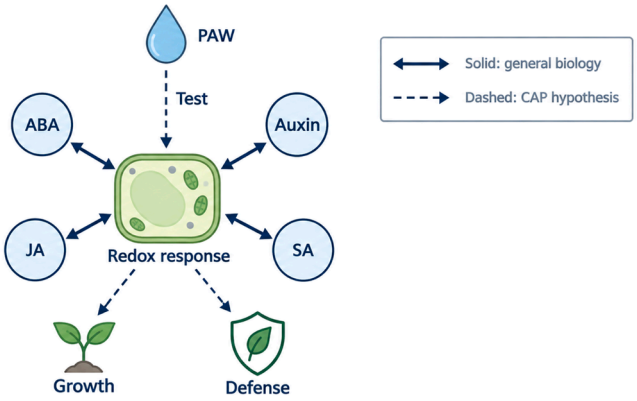


Fig. 4. Testing redox and hormone connections. An original hypothesis diagram separates general redox–hormone interactions from CAP-specific causal links. Hormone measurements and pathway perturbation are required to connect an early response to growth or defense.

and glutaredoxins connect thiol buffering with signal transmission, making pool size, reduction state, and protein targets more informative than total glutathione alone [54]. Organelle-specific hydrogen sulfide metabolism can regulate autophagy and cadmium resilience through redox homeostasis, underscoring the need to resolve compartments [55]. Growth temperature changes the relationship among biomass, photosynthesis, hydrogen peroxide, and glutathione in cucumber, illustrating the environmental dependence of the signaling window [56]. Jasmonate can coordinate the ascorbate–glutathione cycle with the growth–redox continuum under combined drought and sodicity in soybean [57]. We therefore define adaptive buffering by three properties: it limits damaging excursions, preserves a transient signal, and restores a new or original steady state after the stimulus. Superoxide dismutase, catalase, ascorbate peroxidase, glutathione peroxidase, glutathione, and ascorbate should be profiled as a network, ideally with redox ratios, subcellular localization, and flux-sensitive assays. An isolated increase in enzyme activity after CAP may reflect beneficial acclimation, unresolved stress, or compensatory injury. Only the joint trajectory of redox markers and function distinguishes these states (Table 4).

5. Redox–hormone crosstalk triggered by CAP

Hormones do not sit downstream of redox signals in a simple linear cascade. Redox state affects hormone biosynthesis, transport, perception, and transcription, while hormones reshape ROS production, antioxidant capacity, metabolism, and tissue sensitivity. CAP may therefore change hormone output without markedly changing bulk hormone concentration, or it may change concentration without proving that the hormone caused the phenotype. Mechanistic studies need temporal hormone measurements, response reporters, and perturbation of both hormone and redox nodes. (Fig. 4)

5.1. ABA, auxin, and root architecture

Redox and protein acetylation coordinate an ABA-independent osmotic-stress response, showing that stress acclimation can share redox machinery with ABA pathways without being reducible to ABA concentration [64]. Auxin, cytokinin, and ROS interact at sites of primary growth, providing a mechanistic context for CAP-associated changes in root emergence and meristem activity [65]. Ethylene, jasmonate, and ABA jointly regulate salt responses during germination and early squash growth, illustrating why single-hormone measurements can misrepresent a network response [66]. We propose that root phenotypes be decomposed into initiation, emergence, elongation, branching, and nutrient-acquisition components. A larger root system after plasma-activated water could reflect nitrate supply, altered auxin transport, changed cell-wall properties, local peroxide signaling, or compensation for stress. Isotope-labeled nitrogen, auxin reporters, ROS imaging, and cell-type-resolved transcriptomics can discriminate these alternatives. Without that resolution, “hormonal modulation” remains a plausible but weak explanation.

5.2. Ethylene, jasmonate, and salicylate

Salicylate and jasmonate coordinate immune strategy through context-dependent antagonism and cooperation [67]. Their crosstalk can redirect defense investment, thereby altering both pathogen resistance and growth costs [68]. Jasmonate also interacts with secondary metabolism and other growth regulators to maintain redox balance during crop improvement under salt stress [69]. Hormone signaling and metabolism are bidirectionally coupled, so a CAP-driven metabolic shift may, in turn, alter hormonal responsiveness [70]. Plasma-activated water combined with salicylic acid has been reported to improve tomato germination and seedling growth through associated redox and hormonal responses, but the combined treatment does not, by itself, establish the direction of causality [71]. Transgenerational redox priming and hormone–ROS crosstalk have been associated with barley tolerance to combined cold and drought, providing a benchmark for the level of evidence required before durable CAP memory can be claimed [72]. Exogenous jasmonate and salicylate altered maize resistance to fall armyworm, as well as growth and redox homeostasis, illustrating the growth–defense trade-off that CAP studies should quantify [73]. The jasmonate-salicylate interaction alters the plant metabolome and herbivore outcomes, underscoring that immune performance is a network phenotype rather than a single marker [74]. A defensible CAP–hormone model should satisfy intervention logic: block the hormone pathway, retain the redox perturbation, and test whether the phenotype disappears or changes sign. It should also measure fitness costs, because stronger defense-gene expression may coexist with reduced carbon gain or reproduction. The desirable program is not maximum defense; it is the minimum redox input that produces an appropriately timed protective state at an acceptable growth cost.

6. CAP-driven photosynthetic and metabolic reprogramming

6.1. Chloroplasts as sensors, amplifiers, and vulnerable targets

Chloroplast metabolism is governed by thioredoxin and related redox systems that couple light-driven electron flow to enzyme activity [75]. Photosynthetic acclimation under global-warming stresses depends on chloroplast redox signaling and plasticity rather than constitutive ROS suppression [76]. Plants prevent photosynthetic redox imbalance through coordinated photoprotection, alternative electron sinks, antioxidant reactions, and repair [77]. Quantitative proteomics under fluctuating light has revealed redox-dependent functional regulation across photosynthetic machinery [78]. Tissue-resolved HyPer7 imaging can track plastid peroxide dynamics with spatial and temporal resolution sufficient to test CAP-response kinetics [33]. Whole-plant redox imaging in potato demonstrates that organ-scale stress trajectories can be tracked non-destructively rather than inferred from destructive endpoints [79]. CAP may improve photosynthetic performance indirectly by accelerating establishment, improving nutrient status, priming antioxidants, or reducing pathogen load. It may also impair photosynthesis through stomatal closure, pigment oxidation, disruption of electron transport, or carbon-costly defenses. Chlorophyll content alone cannot distinguish among these pathways. Minimum evidence should include gas exchange, PSII operating efficiency and photoinhibition, non-photochemical quenching, pigment state, leaf area, and recovery, normalized to developmental stage.

6.2. Carbon–nitrogen coordination and secondary metabolism

Reactive nitrogen delivered in plasma-activated water may serve simultaneously as a nutrient and a signal. This dual role creates a major confound: better growth after PAW can reflect nitrate fertilization rather than a distinct redox program. The experiment must include mineral-, pH-, conductivity-, and peroxide-matched controls, as well as a complete artificial-PAW control containing the measured stable products. Only residual effects beyond those accounted for by those controls justify invoking emergent chemistry or signaling. Redox regulation of transcription factors links environmental state to stress acclimation and development, providing a route from short-lived redox changes to longer metabolic adjustments [80]. Thioredoxins regulate carbon–nitrogen metabolism through conserved redox mechanisms and are therefore promising nodes for testing whether CAP coordinates assimilation rather than merely supplying nitrogen [81]. Phenylpropanoid, flavonoid, and antioxidant accumulation should be interpreted as flux outcomes that depend on carbon allocation, precursor availability, enzyme regulation, and tissue age. Increased concentration may also result from smaller biomass or dehydration, so absolute content per organ and growth-normalized values should accompany concentration data. The most convincing metabolic study would combine isotope tracing, enzyme redox state, metabolite pools, gas exchange, and biomass allocation. Such a design can determine whether plasma-derived nitrogen enters plant metabolism, whether carbon skeletons are reallocated, and whether secondary metabolites rise through biosynthesis or concentration effects. Redox programming becomes biologically meaningful when it predicts these fluxes across dose and time. CBF transcription factors integrate chloroplast redox state, phytochrome, and hormone signaling during cold acclimation, providing a candidate network motif for testing CAP effects on photosynthetic stress preparedness [82].

The importance of PAW nitrate depends on the counterfactual nutrient supply. Comparing PAW to distilled water in a nitrogen-limited system tests combined nutrition and chemistry; it does not establish benefits beyond conventional fertilizer. Under complete nutrient supply, additional nitrate may have little benefit or may alter root architecture and nitrate accumulation. Report total nitrogen from PAW and all other inputs in mg N $plant^{-1}$ or kg N ha^{-1}, with irrigation volume and timing. Compare equal total N and appropriately matched counterions. Where available, ^{15}N tracing separates plasma-derived nitrogen assimilation from changes in background nitrogen uptake.

7. CAP-induced stress adaptation and biological memory

7.1. Priming is a response property, not a treatment label

Priming means that an initial stimulus changes the response to a subsequent challenge; improved growth immediately after a single treatment is stimulation, not priming. Stress memory adds persistence, implying that the altered response survives a recovery interval and can be retrieved by a subsequent event. Epigenetic memory is narrower still and requires a durable chromatin-associated mechanism rather than persistent metabolites, altered seed hydration, or selection among seedlings. During plant stress, redox signaling interacts with DNA methylation, histone state, chromatin accessibility, and regulatory RNA, creating plausible routes from transient chemistry to persistent transcriptional competence [83]. Redox–epigenetic crosstalk can affect chromatin dynamics through both ROS and RNS, but those general mechanisms do not establish that CAP produces stable epigenetic memory [84]. Rice research distinguishes stress priming from memory and emphasizes the need for defined recovery and rechallenge designs [85]. Salt priming in soybean has been linked to DNA hypomethylation, demonstrating the kind of molecular persistence that a CAP-memory claim would need to document [86]. Arabidopsis salt-memory studies combine epigenetic and transcriptomic features over time, offering a template for distinguishing transient expression from a retained state [87]. Cold-resilient-crop frameworks extend memory to systemic priming and cross-tissue coordination [88]. Contemporary models of plant stress memory integrate metabolic reprogramming, organelle signaling, and systemic acclimation, rather than attributing persistence to a single chromatin mark [89]. Radiation hormesis offers an instructive comparator because a physical stimulus can prime plants via ROS while also causing damage at higher exposure levels [90].

Redox-driven seed priming with silver nanoparticles has been explicitly connected to germination and stress memory, showing that CAP is part of a broader class of externally imposed redox interventions [91]. Quinoa research proposes redox priming as a way to reduce drought adversity, but translation depends on demonstrating benefit under a later, independently imposed stress [92]. Cytokinin sits at the intersection of priming and photosynthetic control, illustrating how hormonal state can preserve or erase the consequences of an initial redox event [93].

7.2. A minimum evidence standard for CAP memory

We recommend four tests. First, include a treatment–recovery–challenge sequence with untreated, treatment-only, challenge-only, and treatment-plus-challenge groups. Second, show that the primed group responds differently after the initial redox and growth markers have substantially returned toward baseline. Third, identify a retained molecular or physiological state and, where possible, perturb it. Fourth, separate within-generation persistence from intergenerational and transgenerational inheritance. Until these criteria are met, “sustained effect” or “persistent priming-like response” is more accurate than “memory.”

Across these mechanisms, the main comparison is between demonstrated CAP responses and established biological routes that could explain them. A transcript increase after PAW has a different evidential status from a sensor response, and both differ from a loss-of-function test. General redox literature provides plausible nodes, while plasma-specific studies must establish delivery and temporal order. The molecular and application evidence tables retain this distinction so that mechanistic depth is not inferred simply from the number of omics measurements.

8. Agricultural applications of CAP-mediated redox modulation

8.1. Seed technology: from surface engineering to establishment

Seed technology combines a well-defined pretreatment opportunity with a particularly strong surface-hydration confound. General seed-redox studies identify plausible endogenous routes [94–97], but a CAP treatment also changes the conditions of imbibition. Potato and paddy crop studies extend follow-up beyond germination [98,99]; controlled peanut hydroponics provides another defined delivery system [100]. The mechanistic question is whether early chemistry, surface change or supplied nutrients explain a later phenotype. Artificial and neutralized PAW comparisons are therefore evaluated separately in Section 8.5 rather than cited merely as favorable examples.

Soybean responses depend on the CAP regime [101], while lettuce substrate-volume experiments show that root-zone context changes how we interpret PAW treatment [22]. Seed immersion in PAW shows that discharge need not directly contact the seed [102]; Ferula argon-plasma priming illustrates a separate direct route [103]. These studies address different transport and surface conditions and do not establish one common dose. An appropriate comparison must retain species, seed lot, hydration, delivery route, and contact chemistry rather than rank treatment durations from unrelated devices.

Combined seed-and-water treatment [104] and broccoli sprout studies [105] add distinct establishment and quality endpoints. The agronomic target remains successful establishment, not an isolated germination percentage. Earlier radicle protrusion accompanied by abnormal seedlings, unfavorable allocation or poorer later growth cannot define success. Multi-lot studies should quantify emergence timing, soil establishment, uniformity, and reproductive carryover. Seed-coat properties, storage history, microbiota and baseline viability modify both accessibility and buffering, making lot-aware calibration essential.

8.2. Abiotic-stress management

PAW studies differ in whether they test an imposed challenge, nutrient supply, or general growth promotion. Kentucky bluegrass seed priming and irrigation addressed drought resistance [106], whereas maize landraces highlighted genotype-dependent responses under abiotic stress [107]. Barley foliar treatment addressed distal root growth [108]. These outcomes require different denominators: challenge protection, water use, root exploration, and developmental acceleration should not be combined into a single “stress tolerance” effect.

Wheat physiology and antioxidant-enzyme studies offer multi-endpoint association [109], while maize–barley comparisons demonstrate species-dependent responses to the same broad treatment modality [110]. Agreement among pigments, biomass, and enzyme activities can strengthen biological coherence, but it does not identify the initiating redox species. The strongest comparison uses measured contact chemistry, equal nutrients and a treatment-by-stress design with matched developmental stage and recovery.

Abiotic-stress benefit must be claimed relative to the correct denominator. Better performance under drought may arise because CAP accelerated early development, increased root exploration, altered stomatal behavior, supplied nitrogen, or created genuine stress preparedness. A mechanistic design compares well-watered and stressed plants at matched developmental stage, quantifies water use, and tests recovery. For salinity and heavy metals, ion accumulation, exclusion, sequestration, antioxidant state, and growth dilution must be distinguished. For heat and cold, membrane stability and post-stress regrowth are more informative than an increase in enzyme activity measured during the challenge. Three types of PAW have produced different lettuce-germination and growth outcomes, confirming that “PAW” cannot be treated as a uniform input [111]. Combined non-thermal plasma and PAW treatments affected water uptake and early growth across diverse crop seeds, suggesting both shared and species-specific components of the response [112]. Plasma-linked local optimization requires trait-specific criteria and prospective validation [113]. Red-clover germination and biomass responded to cold atmospheric-pressure plasma, reinforcing the need to include forage and perennial life histories in translation studies [114].

8.3. Disease management: direct inactivation versus induced resistance

CAP can reduce pathogen burden directly through oxidative, nitrosative, electrical, UV-associated, and surface effects, while the treated plant may also activate defense. These routes have different dose optima: a microbicidal treatment can exceed the plant signaling window, whereas a mild plant-priming treatment may not sanitize the surface. Experimental designs must therefore quantify pathogen survival before inoculation, disease progression after washing or transfer, host defense markers, and plant injury. Researchers have tested PAW generated by non-thermal atmospheric plasma against phytopathogenic bacteria, supporting direct antimicrobial deployment in crop systems [115]. Researchers have tested pre-sowing CAP to control winter-wheat seedling blights, directly linking seed sanitation to disease outcomes [116]. PAW can trigger plant defense responses after the liquid is generated, supporting a host-mediated route alongside direct antimicrobial chemistry [117]. Combining CAP with plant secondary metabolites can strengthen antimicrobial activity, but crop-surface safety and interaction mechanisms remain application-specific [118]. The same seed-treatment experiment observed surface-microorganism inactivation and increased germination, illustrating the dual sanitation–establishment objective [119]. Ultralong-lasting PAW inhibited pathogens and improved resistance in soft-rot-infected hydroponic lettuce, a case in which residual chemistry and host defense require careful separation [120]. AP2/ERF transcription factors regulate multiple plant-disease-resistance outputs, offering candidate nodes for linking CAP-induced redox signals to defense transcription [121]. The strongest induced-resistance study would apply CAP before pathogen arrival, remove residual antimicrobial chemistry, challenge with a quantified inoculum, and test defense-pathway mutants or inhibitors. It would also report growth cost and microbiome effects. Sanitization and immunity may be combined operationally, but they should remain analytically distinct.

8.4. Nutrient delivery, crop quality, and productivity

A global meta-analysis found broad but heterogeneous agricultural responses to PAW, so the mean benefit should not be mistaken for a transferable recipe [122]. Current syntheses characterize PAW as promising for sustainable agriculture while emphasizing unresolved fundamentals behind variable outcomes [123]. Broader assessments of PAW across agriculture, food safety, and healthcare show the chemistry's translational reach but also the need for application-specific specifications [124]. Recent studies report multifaceted PAW effects on germination and plant development, reinforcing the need for response-specific rather than product-wide claims [125]. Interfacial modulation of nitrogen uptake has been proposed as a mechanism by which PAW promotes root growth, a hypothesis that can be tested with isotope tracing and transporter perturbation [126]. CAP increased oat germination, photosynthetic pigments, and antioxidant enzyme activity, but these correlated outcomes do not yet establish a single causal redox pathway [127]. PAW improved potato growth, physiology, and yield in association with RONS, motivating season-scale studies that include storage quality and resource efficiency [128]. PAW used as an aeroponic growth medium altered germination, physiology, and biochemistry, a configuration well-suited to continuous chemical monitoring [129]. DBD-activated water affected hydroponic lettuce growth, illustrating the operational compatibility of plasma liquids with controlled-environment production [130]. Pin-to-plane DBD has been

Table 5
Selected plant studies with mechanistic evidence and application readiness.

Study	Design and endpoint	Evidence interpretation	Grade / readiness
Radish [137]	Seed-surface chemistry; germination	Surface route; reproductive follow-up missing	C / R0
Mung bean [138]	Direct CAP; surface and growth	Etching/imbibition contribution unresolved	C / R0
Spinach [139]	Persistent germination response	Persistence is not a causal memory mechanism	C / R0
Agropyron [140]	Direct plasma plus PAW; damage	Combined exposure does not isolate redox action	C / R0
Prosopis [141]	CAP seed treatment under salinity	Controlled stress phenotype; crop transfer untested	C / R0
Mung bean [142]	Seed CAP with field follow-up	Field endpoint; independent environments needed	C / R1
Papaya [143]	CAP and transcriptomic shifts	Association; no complete causal chain	B / R0
Soybean [144]	CAP; antioxidant and growth endpoints	Enzyme responses do not identify a sensor	C / R0
Aged pepper [145]	PAW; germination and growth	Seed-lot condition may modify response	C / R0
Tomato [18]	PAW and nitrate comparator; crop cycle	Nutrient discrimination; pathway causality incomplete	C / R1
Radish [146]	PAW nitrogen source in hydroponics	Nutritional use case; energy economics unresolved	C / R1
Rice [147]	CAP; plant–herbivore bioassay	Induced phenotype; redox causality incomplete	C / R0
Soybean and sorghum [148]	CAP; growth and herbivore response	Species-specific optima; multi-environment tests needed	C / R0
Wheat [11]	Artificial PAW versus plasma PAW	Stable-mixture sufficiency; absence of difference is not equivalence	C / R0
Barley [149]	Acidic versus Mg-neutralized PAW	pH and Mg chemistry are coupled	C / R0
Lettuce [22]	PAW and substrate-volume comparison	Buffering and nutrient context modify endpoints	C / R1
Sorghum [61]	PAW dose series; RNA sequencing	Measured chemistry; no nitrate-matched treatment	B / R0
Mung bean [14]	Air/N_2 PAW; transcriptome and qPCR	Chemistry plus molecular response; causality incomplete	B / R0
Plant cells [10]	CAP species; live-cell Ca^{2+}/peroxide	Early signal evidence; no agronomic chain	B / R0
Arabidopsis [60]	CAP-derived RONS and redox state	Chemistry–biology linkage; recovery and field chain incomplete	B / R0

coupled with selective PAW production for microgreen cultivation and microbial safety, showing how growth and sanitation objectives can be co-designed [131].

PAW foliar application has increased rice growth and yield, but researchers must resolve foliar nutrient and redox contributions with matched solutions [132]. PAW improved growth and nutrient uptake in maize and pea, supporting simultaneous measurement of ion acquisition and signaling markers [133]. Gliding-arc treatment of agricultural wastewater produced PAW, which researchers then tested in growth media and crop production, illustrating a circular resource-use case with substantial matrix complexity [16]. PAW affected both vegetative and reproductive growth of Conyza canadensis, showing why crop-cycle outcomes should follow seedling observations [134]. Combined LPGOD plasma and PAW treatment affected rice, adding another example in which direct and indirect components require factorial separation [135]. Researchers have investigated alkaline PAW as a wheat nutrient input, showing that pH direction and nitrogen chemistry can be engineered in tandem [136]. Crop-quality claims need compositional depth and exposure realism. Higher levels of phenolics or antioxidants may be beneficial, but sensory quality, anti-nutrients, shelf life, residue chemistry, and yield dilution must also be measured. Productivity claims require randomized, replicated field trials across environments and input inventories, compared with best conventional practice. Proceed with a treatment only if the added device and energy costs are justified by a stable gain in yield, quality, input efficiency, or risk reduction.

8.5. *Comparative synthesis and chemistry-control audit*

Table 5 compares selected studies by endpoint, chemical discrimination, mechanistic grade and application readiness. The rows are an evidence map rather than a pooled effect estimate. Cross-study differences in crop, source, baseline nutrition, treatment route and outcome prevent a single dose–effect ranking. Within-study matched comparisons are more informative than a catalog of positive percentages. For mechanistic inference, artificial-PAW sufficiency and nitrate-matched comparisons provide stronger discrimination than antioxidant-enzyme correlations; for deployment, reproductive endpoints provide stronger follow-up than germination alone.

A focused audit of five PAW studies examined the treatment comparators in the accessible methods: wheat artificial PAW [11], barley Mg-neutralized PAW [149], lettuce substrate-volume treatments [22], sorghum transcriptomic treatments [61], and tomato nitrogen-replacement treatments [18]. Three of these five included an artificial mixture, a neutralization comparison or an explicit nitrate comparator. One used artificial PAW; one used Mg-mediated neutralization; one included an explicit nitrate comparator (tomato). Sorghum used pH-adjusted PAW without a separate nitrate-matched treatment. Mg-neutralized PAW is a modified treatment, not an isolated pH-only control, and an artificial mixture is not automatically a complete species-and-kinetics match. No whole-corpus frequency is inferred from this targeted denominator.

In barley, Mg-neutralized PAW improved early germination and seedling endpoints relative to acidic PAW, including greater pigment and soluble-protein responses [149]. This is evidence that treatment formulation matters, but magnesium supply and chemical changes accompanying neutralization prevent attribution to pH alone. A further experiment should match magnesium, ionic strength, and post-neutralization peroxide/nitrite/nitrate, alongside a separately acidified untreated-water control. The comparison strengthens the case for chemistry-aware design without establishing that neutralized PAW always occupies a superior redox window.

The wheat artificial-mixture comparison found no consistent statistical superiority of plasma-generated water for the principal growth responses [11]. In the same dataset, shoot dry mass increased while root length decreased; therefore, a positive shoot endpoint alone cannot define agronomic benefit. Failure to detect a difference is not a formal equivalence test. Tomato nitrogen comparisons similarly support assessing PAW against equal-N fertilizer rather than only water [18]. Collectively, these results make chemical sufficiency a central test of the framework, rather than an inconvenient exception.

Descriptive contrasts calculated from the published wheat means were +6.66 percentage points for germination, +38.4% for shoot dry mass and −37.0% for root length with plasma PAW versus water [11]. These summarize direction and magnitude without adding a new inferential test. They are not pooled effects and do not establish superiority over artificial PAW. In tomato, the reported mature-biomass ratios were 3.9 for PAW400 and 3.1 for nitrate fertilizer relative to water; that water baseline was nutrient-limited, and equal-N comparison remains the more relevant agronomic counterfactual [18].

If a rigorously matched chemical mixture reproduces the phenotype,

Table 6
Plasma-focused molecular measurement and validation designs.

Plasma-specific study	Measurements	Mechanistic value and required follow-up	Reference
Arabidopsis leaf CAP	Early stress-gene expression	Establishes timing; add measured interface chemistry and sensor perturbation	[58]
Arabidopsis seed CAP	Comparative transcriptome	Response modules; add stage matching, reporter dynamics and complementation	[59]
Mung bean PAW	Chemistry, RNA sequencing and qPCR	Links formulation to response; test matched mixtures and candidate genes	[14]
Sorghum PAW	Water chemistry, transcriptome, histology and biomass	Multi-level association; add equal-N controls and causal validation	[61]
Plant cells and CAP species	Time-resolved intracellular Ca^{2+} and peroxide	Early response; link to later phenotype and pathway-specific perturbation	[10]

there is no demonstrated plasma-specific biological advantage for that endpoint. Plasma could still have an operational advantage through on-site generation, coupled sanitation or reduced transport of reagents; each advantage requires evidence. A comparative cost boundary should include capital amortization, electricity, feed gas, water conditioning, cooling, maintenance, storage losses, labor and chemical assay costs for plasma, and procurement, transport, dilution, storage, dosing and equivalent verification for the chemical alternative. Report cost per viable seed treated, per hectare or per kilogram of marketable product, and kWh per mole of fixed nitrogen where nutrient delivery is the purpose.

A fair comparison uses solutions that meet the same chemical specification at plant contact and reach the same biological endpoint under the same background management. It then estimates incremental net return, not merely energy use per litre of PAW. Until replicated economic and lifecycle analyses show otherwise, lower chemical procurement does not establish lower total cost or lower environmental impact. The framework supports selecting the simpler chemical route where it provides equivalent performance, and reserving plasma for use cases with demonstrated added process value.

9. Integrating multi-omics to decode plasma-induced responses

9.1. From parallel lists to causal layers

Organize transcriptomics, proteomics, metabolomics, and redoxomics around plasma-specific chemical exposure and shared sampling times. The aim is to connect a measured input to early signaling, later molecular adjustment and a prospective phenotype. Generic digital-agriculture examples do not establish that connection. Table 6 therefore prioritizes CAP and PAW studies, and Fig. 5 presents a proposed measurement-to-validation workflow rather than a demonstrated autonomous technology.

9.2. Experimental design for identifiable mechanisms

At minimum, omics designs require untreated controls, artificial PAW, pH and nutrient comparators, a beneficial dose, an injurious dose, and early–late sampling. Surface, vascular, meristematic and chloroplast compartments may receive different inputs. Rapid quenching is needed to preserve protein redox modifications during sampling. Candidate pathways should be tested in an independent perturbation or validation cohort, not by enrichment analysis alone.

Table 6 prioritizes plasma-specific transcriptomic, physiological, and live-cell studies. It separates what each experiment measured from the additional experiment needed to build a causal chain. Transcriptome abundance cannot establish a cysteine switch, and a metabolite pool cannot establish flux. A useful integration starts with matched chemistry and sampling times, then asks whether an early molecular response predicts a later phenotype across independent batches. Candidate pathways should be followed with targeted modification analysis, mutant/complementation tests or a carefully controlled perturbation. This is a plasma-focused research design rather than a claim that generic multi-omics already makes treatment predictive.

10. Toward precision redox agriculture

10.1. Closed-loop plasma biotechnology

We propose a local feedback development sequence: measure, model, treat, validate and update. Measurement characterizes the source, contact chemistry and baseline plant state. A model predicts a target response for one crop–reactor–stage combination. Independent validation evaluates that prediction against predefined chemistry,

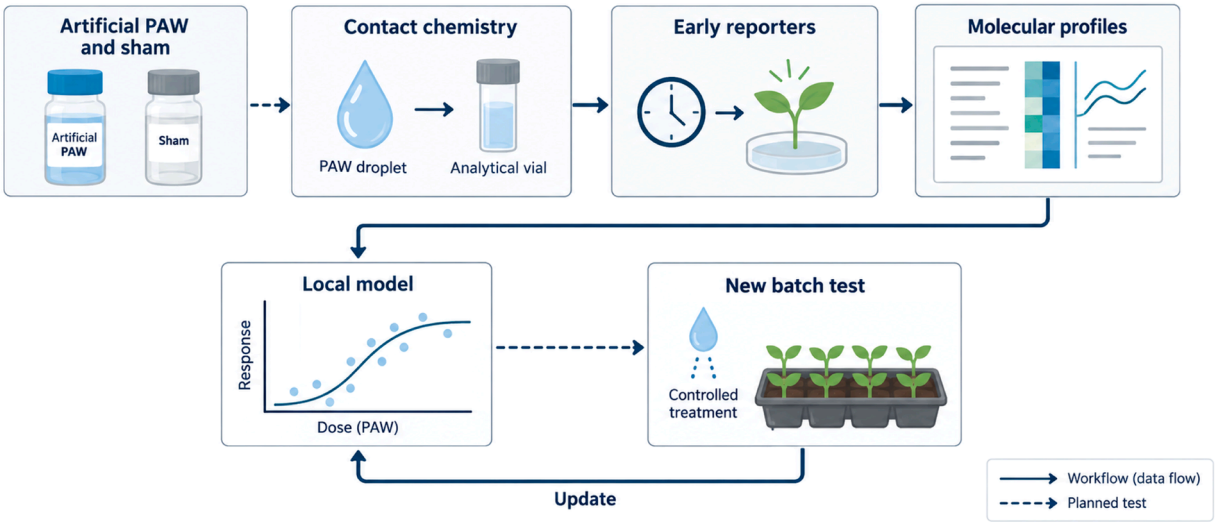


Fig. 5. A plasma-focused measurement and validation workflow. An original proposed workflow connects measured chemistry and timed redox, transcript and metabolite sampling to local prediction and independent validation. It is a development plan, not a demonstrated autonomous precision platform.

recovery and phenotype criteria. Only observations from a new experiment should update the treatment rule. This is a proposed route to programmability, not evidence that the field already has a general precision platform.

A concrete local optimization example is the wheat PAW haloprimming study of Asghari et al. [113]. Its response-surface design varied water activation level, priming duration and salinity, and predicted trait-specific optima. The germination optimum was 0.13 min mL^{-1}, 10.3 mM NaCl and 120.22 min priming, whereas fresh-weight optimization selected different conditions. The reported discrepancy between adjusted and predicted model performance reinforces the need for independent validation. This is a plasma-linked local optimization example, not an implemented real-time closed loop.

To extend that design prospectively, one wheat lot and one reactor would be calibrated first. Replace activation time per volume with measured contact peroxide, nitrate, nitrite and pH, while retaining priming duration and salinity. Randomize independent PAW batches, reserve an unseen seed lot, and predefine germination speed and abnormal-seedling frequency alongside biomass. A constrained response-surface or Bayesian model selects the next treatment only within the tested chemical domain. Confirmation must reproduce the predicted gain and recovery profile against sham and artificial-PAW controls. A verified update from new observations would then constitute a local feedback step; cross-crop precision remains a further validation task.

10.2. Inputs, outputs, and decision rules

The input space should include absorbed power, waveform, gas, humidity, distance, treated geometry, liquid composition, species concentrations, storage time, genotype, seed lot, developmental stage, environment, and treatment history. The output space should include early redox and calcium trajectories, molecular markers, growth, stress performance, yield, quality, injury, and resource efficiency. Safety constraints belong inside the optimizer: maximum tissue injury, undesirable nitrate or nitrite accumulation, off-target microbiome change, operator exposure, and energy intensity. The first useful models will be local rather than universal. Bayesian optimization or response-surface methods can learn an efficient treatment rule for one reactor–crop–-stage combination while preserving uncertainty. Hierarchical models can then ask which parameters transfer across genotypes, seasons, and devices. Model performance should be judged prospectively on unseen batches and environments; retrospective fit is insufficient evidence of programmability.

11. Limitations, reproducibility, and future challenges

11.1. Reporting and controls

The field's largest obstacle is not a lack of positive observations but an inability to compare them. Device names hide operating regimes, and exposure time hides the delivered chemistry. Every study should report electrical diagnostics, gas conditions, temperature, geometry, environmental humidity, target properties, treatment uniformity, and a defined panel of reactive products. For liquids, pH, conductivity, oxidation–reduction potential, temperature, nitrate, nitrite, hydrogen peroxide, storage conditions, and delay to use are minimum requirements rather than complete specifications. Plant-side metadata are equally important: genotype, seed source and age, storage, initial moisture, dormancy status, microbiological status, developmental stage, growth environment, time of day, and baseline nutrition. Controls should be chosen to answer mechanisms, not just to satisfy convention. Sham exposure controls handle and gas flow; pH-, conductivity-, temperature-, nitrate-, nitrite-, peroxide-, and mineral-matched controls decompose stable chemistry; artificial PAW tests sufficiency; and antioxidant or pathway perturbations test causality. A green-light-excitable FRET system can monitor intracellular calcium in plant cells while reducing spectral conflict with common reporters, illustrating the sensor engineering needed for real-time CAP mechanistic studies [48]. Such sensors should be paired with instrument synchronization so that plasma pulses, chemical measurements, and biological trajectories share a clock. Open raw waveforms, calibration curves, code, and negative results would materially improve comparability.

Analytical specificity limits constrain every mechanistic conclusion. ORP is a mixed electrode potential affected by pH, temperature, reference electrode, and redox couples; it cannot be converted into a RONS concentration. Test strips are useful for screening but require quantitative confirmation and a clearly defined detection range. DCFH-DA oxidation is not specific for peroxide or superoxide and is influenced by peroxidases, metals, and probe-derived chemistry; it should be described as a nonspecific oxidation readout rather than a species-resolved ROS assay [150]. DAF-derived NO readouts likewise require controls for uptake, oxygen dependence, and interfering chemistry [42].

For stable aqueous species, use matrix-calibrated peroxide photometry or a validated enzymatic assay with catalase and spike-recovery controls; Griess-based nitrite analysis with blanks and interference testing; and ion chromatography for separate nitrate and nitrite quantification. Acid–base titration cannot uniquely assign nitrate in a solution containing other acids. Short-lived radical studies need validated trapping and, where appropriate, electron paramagnetic resonance, recognizing that trapping changes the reaction system. $ONOOH/ONOO^-$ inference should combine kinetic precursor measurements and orthogonal product analysis; a single nonspecific fluorescent signal is insufficient. State calibration range, limit of detection, limit of quantification, recovery, technical precision, and biological sampling unit.

11.2. Scale, ecology, and safety

Laboratory uniformity will not survive unchanged in trays, seed-treatment lines, irrigation networks, greenhouses, or fields. Larger reactors introduce residence-time distributions, electrode fouling, gas-flow heterogeneity, and energy-transfer losses. Agricultural water introduces bicarbonate, chloride, metals, particles, and organic matter that reshape reactive chemistry. Soil and rhizosphere communities can consume products, produce new metabolites, and alter plant access. Scale-up must therefore preserve a verified chemical and biological specification, not merely geometric similarity. Long-term assessment should include reproduction, seed quality, beneficial microbiota, non-target organisms, nitrogen emissions, water chemistry, worker exposure, and lifecycle energy. CAP may reduce pesticide or fertilizer use in some use cases, but that environmental benefit must be demonstrated through system boundaries and counterfactuals. A credible commercialization claim needs quality-control tolerances, failure modes, maintenance requirements, economics, and regulatory classification for the intended jurisdiction.

11.3. Evidence grading

Mechanistic evidence and production validation require separate assessments. Table 5 applies the following claim-specific criteria to selected studies. It does not rank publications by phenotype size or analytical platform complexity.

Application readiness is scored separately: R0 denotes mechanistic or laboratory feasibility without a production endpoint; R1 denotes controlled crop or greenhouse follow-up including a relevant growth, quality or reproductive endpoint; R2 requires independently replicated production trials across environments, against standard management and with input accounting; R3 requires a reproducible operating specification, economics, lifecycle assessment and deployable quality control. A study can therefore be B/R0 or C/R1. Mechanistic grades must never be assigned to pipeline stages, and a proposed validation experiment cannot be labeled Grade A before it is performed (Fig. 6).

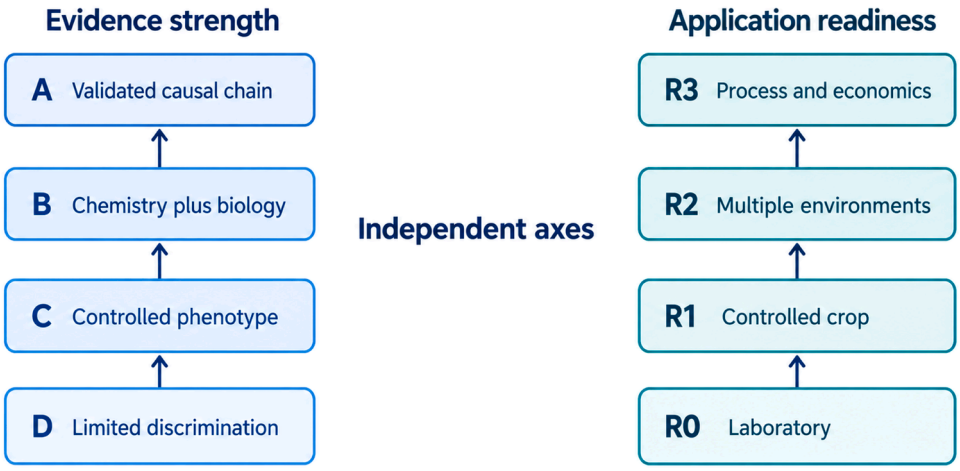


Fig. 6. Independent evidence and application readiness. An original framework presents mechanistic evidence grades A–D separately from application readiness R0–R3. The categories evaluate evidence already obtained; they are not labels assigned to successive pipeline operations or approvals of technology.

For conservative grading, A requires the complete chemistry–signal–causal perturbation–recovery–agronomic chain with independent-condition validation. B requires characterized chemical input linked to molecular and physiological responses but lacks that complete chain. C supports a controlled phenotype with incomplete chemical or causal discrimination. D indicates that the claimed redox interpretation relies primarily on exposure time, an isolated marker, or poorly discriminating controls. Grades refer to the stated claim and accessible evidence, not journal prestige or overall study quality. Selected examples in Table 5 do not establish Grade A programmable agriculture; the matrix identifies the missing link needed to test that claim.

12. Conclusions

Cold atmospheric plasma should be understood as a potential external redox-signaling technology, not as a universally beneficial stress generator. Its distinctive opportunity lies in controllable timing, adjustable chemistry, direct or water-mediated delivery, and compatibility with non-genetic crop management. Its central risk lies in the same properties: complex mixtures, steep spatial gradients, context-dependent responses, and a narrow boundary between adaptive signaling and injury. The redox-programming framework replaces the question "How long was the plant treated?" with four questions: what chemical burden arrived, on what timescale, at which responsive tissue, and against what buffering state? Biological redox dose provides the measurement scaffold, while closed-loop plasma biotechnology provides the development pathway. Success will be demonstrated when a treatment selected from those variables prospectively produces a defined molecular trajectory and agronomic outcome across independent conditions. Until then, CAP is best described as a promising, mechanistically testable redox input rather than a finished precision technology.

CRediT authorship contribution statement

Mojtaba Kordrostami: Writing – review & editing, Writing – original draft, Supervision. **Abolfazl Mazandarani:** Formal analysis, Conceptualization. **Ali Ebadi:** Software, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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